Customer Behavior Analysis Dashboard – Project Documentation

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Programme: MCA Semester I, SPPU (ZIBACAR, Pune), 2026-27 **Tools:** Excel · Python (Pandas) · MySQL · Power BI

1. Project Overview

This project analyses customer shopping behaviour for a retail business to understand **who buys, what they buy, how much they spend and what drives repeat purchases**. Raw transaction data is cleaned and feature-engineered in Python, loaded into MySQL, queried with 10 business questions in SQL, and finally presented in an interactive Power BI dashboard.

Objectives

- Clean and prepare the customer shopping dataset for analysis.
- Answer key business questions on revenue, discounts, subscriptions, shipping, products and customer loyalty.
- Build an interactive dashboard so stakeholders can filter and explore the data themselves.
- Give practical recommendations from the findings.

2. Project Files

File	Purpose
customer_shopping_behavior.csv	Raw dataset (3,900 rows × 18 columns)
customer_shopping_behavior.ipynb	Python notebook: data loading, cleaning, feature engineering, load to MySQL
customer_behavior_analysis_dashboard.sql	10 SQL business-analysis queries
customer_behavior_analysis_dashboard.pbix	Power BI dashboard (1 page)

3. Workflow / Architecture

CSV → Python (Pandas) → MySQL (database `customer_behavior`, table `customer`) → SQL analysis → Power BI dashboard

1. **Data understanding:** `head()`, `info()`, `describe()`, null check.
2. **Cleaning and feature engineering in Python.**
3. **Load to MySQL** using SQLAlchemy and PyMySQL (`df.to_sql`).
4. **Analysis in SQL** (aggregations, CASE, subqueries, CTEs, window functions).
5. **Visualisation in Power BI** connected to the `customer` table.

4. Dataset Description

3,900 customer purchase records, 18 columns, 50 US states (locations) and 25 distinct products.

Column	Description
Customer ID	Unique customer identifier (1–3900)
Age	Customer age (18–70)
Gender	Male / Female
Item Purchased	Product name (25 products)
Category	Clothing, Accessories, Footwear, Outerwear
Purchase Amount (USD)	Order value, $20–$100
Location	US state
Size, Color	Product size and colour
Season	Season of purchase (Spring, Summer, Fall, Winter)
Review Rating	Customer rating (about 2.5–5.0)
Subscription Status	Yes / No
Shipping Type	Free Shipping, Standard, Store Pickup, Next Day Air, Express, 2-Day Shipping
Discount Applied	Yes / No

Column	Description
Promo Code Used	Yes / No
Previous Purchases	Number of earlier purchases
Payment Method	PayPal, Credit Card, Cash, Debit Card, Venmo, Bank Transfer
Frequency of Purchases	Weekly, Bi-Weekly, Fortnightly, Monthly, Quarterly, Every 3 Months, Annually

Headline numbers: total revenue ≈ **$233,081**, average purchase ≈ **$59.76**, average rating ≈ **3.75**.

5. Data Preparation (Python)

Step	What was done	Why
Column renaming	Lower-cased, spaces replaced with underscores; `purchase_amount_(usd)` renamed to `purchase_amount`	SQL-friendly, consistent names
Missing values	`review_rating` had 37 nulls; filled with the **median rating of the same category**	Keeps data size, avoids skew from outliers
Age group	New column `age_group` using `pd.qcut(age, q=4)` with labels Young Adult, Adult, Middle-aged, Senior	Quartile-based segments of roughly equal size (about 940–1,030 customers each)
Purchase frequency	New column `purchase_frequency_days` mapping frequency text to days (Weekly = 7, Fortnightly/Bi-Weekly = 14, Monthly = 30, Quarterly/Every 3 Months = 90, Annually = 365)	Numeric version of frequency for analysis
Load to MySQL	Table `customer` created in database `customer_behavior`	Enables SQL analysis and Power BI connection

6. SQL Analysis

Each query below answers one business question on the `customer` table.

#	Business question	Technique	Key result
Q1	Total revenue by gender	`GROUP BY` + `SUM`	Male **$157,890** vs Female **$75,191**
Q2	Customers who used a discount but still spent at least the average purchase amount	Subquery with `AVG`	**839** customers (average = $59.76)
Q3	Top 5 products by average review rating	`AVG`, `ORDER BY`, `LIMIT`	Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Handbag 3.78
Q4	Average spend: Standard vs Express shipping	Filtered `GROUP BY`	Express **$60.48** vs Standard **$58.46**
Q5	Do subscribers spend more?	Multiple aggregates	Non-subscribers: 2,847 customers, avg $59.87, revenue $170,436. Subscribers: 1,053 customers, avg $59.49, revenue $62,645
Q6	Top 5 products by discount rate	`CASE WHEN` inside `SUM`	Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37%
Q7	Customer segments: New, Returning, Loyal	CTE + `CASE`	Loyal 3,116, Returning 701, New 83
Q8	Top 3 most purchased products in each category	CTE + `ROW_NUMBER() OVER (PARTITION BY ...)`	Ranked list per category (see query output)
Q9	Are repeat buyers (more than 5 purchases) likely to subscribe?	Filtered `GROUP BY`	2,518 non-subscribers vs 958 subscribers

#	Business question	Technique	Key result
Q10	Revenue by age group	`GROUP BY` + `ORDER BY`	Young Adult $62,143, Middle-aged $59,197, Adult $55,978, Senior $55,763

Segment rules used in Q7: New = 1 previous purchase, Returning = 2 to 10, Loyal = more than 10.

7. Power BI Dashboard

The dashboard has one page, built on the MySQL `customer` table.

KPI cards: Number of Customers · Average Purchase Amount · Average Review Rating

Charts

Visual	Type	What it shows
% of customers by Subscription Status	Donut chart	Subscriber vs non-subscriber share
Sales by Age Group	Clustered bar	Orders per age group
Revenue by Age Group	Clustered bar	Age-group comparison
Revenue by Category	Clustered column	Revenue for each product category
Sales by Category	Clustered column	Orders per category

Slicers (filters): Subscription Status · Gender · Category · Shipping Type

All visuals respond to the slicers, so a user can, for example, view only female subscribers buying Footwear with Express shipping.

8. Key Insights

- **Gender:** Male customers generate about 68% of revenue (and make up about 68% of customers, 2,652 of 3,900), so the gap is mostly a customer-count gap, not a spending gap.
- **Category:** Clothing is the largest revenue driver ($104,264), followed by Accessories ($74,200), Footwear ($36,093) and Outerwear ($18,524).
- **Subscription:** Subscribers do **not** spend more per order (about $59.49 vs $59.87). Only 27% of customers are subscribed, so the subscription programme is underused.
- **Loyalty:** About 80% of customers are in the Loyal segment (more than 10 previous purchases), but even among repeat buyers about 73% are not subscribed. This is a clear conversion opportunity.
- **Shipping:** Express customers spend slightly more than Standard (about $2 per order), a small difference.
- **Discounts:** Around half of all purchases of Hat, Sneakers and Coat use a discount, and 839 discounted orders were still above average spend. Discounts are not always needed to drive large orders.
- **Age:** Revenue is spread fairly evenly across the four age groups, with Young Adults slightly ahead.

9. Recommendations

1. Launch targeted subscription offers to Loyal, non-subscribed customers.
2. Review discount depth on products already discounted about 50% of the time, since many big spenders buy without needing one.
3. Run campaigns to attract more female customers, who are under-represented.
4. Promote Clothing and Accessories in peak seasons; push Outerwear bundles to raise its low share.
5. Highlight top-rated products (Gloves, Sandals, Boots) in marketing.

10. Known Issues and Suggested Improvements

These are things worth fixing before final submission or publishing on GitHub.

- **Hard-coded database password:** the notebook contains the MySQL root password in plain text. Remove it (use an environment variable or `getpass`) and change the password before pushing to GitHub.

- **Power BI measures:** "Sales by Age Group" and "Sales by Category" use `Sum(customer_id)`, which adds up ID numbers and is meaningless. Use `Count of customer_id` instead. "Revenue by Age Group" uses `Sum(previous_purchases)`; it should use `Sum(purchase_amount)` to match its title.
- **Notebook cleanup:** `df.describe()`, `df.head()` and the frequency-mapping check cells have no saved outputs, so rerun and save the notebook with outputs.
- **Age-group labels:** Power BI sorts the labels alphabetically; set a custom sort order (Young Adult, Adult, Middle-aged, Senior).
- **Q2 wording:** the query uses `>=` average, so it counts customers at or above average, not strictly above.
- **Extra analysis ideas:** revenue by season, location (state) map, payment-method preference, promo-code impact.

11. How to Reproduce

1. Install the libraries: `pip install pandas sqlalchemy pymysql`.
2. Start MySQL and run `create database customer_behavior;`.
3. Run the notebook cells in order. This cleans the data and creates the `customer` table.
4. Run the queries in `customer_behavior_analysis_dashboard.sql` in MySQL Workbench.
5. Open the `.pbix` file in Power BI Desktop and refresh the data source to point to your MySQL server.

12. Conclusion

The project covers the full analytics pipeline from raw CSV to an interactive dashboard. The main finding is that revenue depends mainly on customer volume and product category rather than on subscription or shipping choice, and that the large loyal base is the biggest growth opportunity through subscription conversion.